

CMSC5743 Lab 01: GEMM

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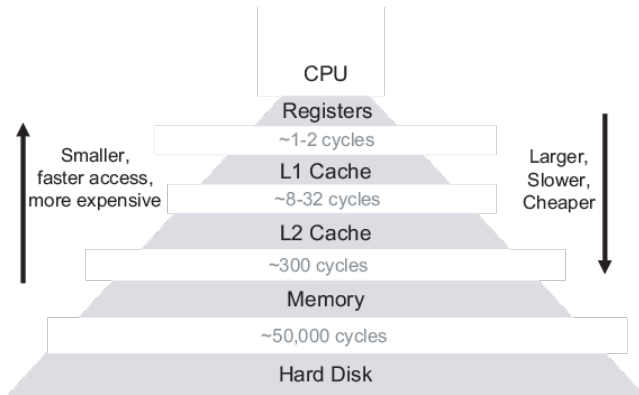
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Memory Hierarchy

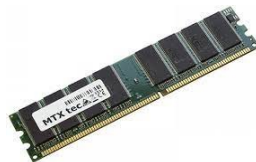


- Memory is primarily of three types :
 - Cache Memory
 - Primary Memory / Main Memory
 - Secondary Memory

- Cache Memory
 - Cache memory is faster than main memory
 - Less access time as compared to main memory
 - Stores the program that can be executed within a short period of time
 - Stores data for temporary use



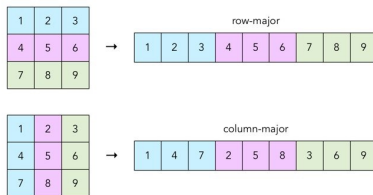
- However ...
 - Cache memory has limited capacity
 - It is very expensive
- Primary Memory (Main Memory):
 - Usually volatile memory
 - Working memory of the computer
 - Faster than secondary memories
 - A computer cannot run without the primary memory



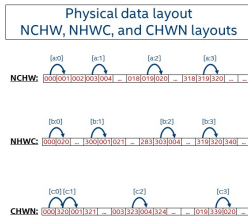
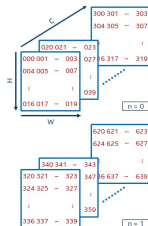
- If the processor finds that the memory location is in the cache, a **cache hit** has occurred and data is read from cache
- If the processor **does not** find the memory location in the cache, a **cache miss** has occurred. For a cache miss, the cache allocates a new entry and copies in data from main memory, then the request is fulfilled from the contents of the cache
- Hit ratio = $\text{hit} / (\text{hit} + \text{miss}) = \text{no. of hits} / \text{total accesses}$

Data Layout

- Matrix:

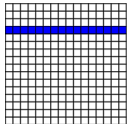


- Tensor:

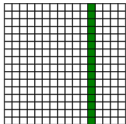


Matrix Multiplication

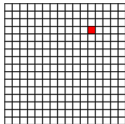
A



B



C



```
// ***** C = A x B *****  
void matmul() {  
    memset(C, 0, sizeof(C));  
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < n; j++) {  
            for (int k = 0; k < n; k++) {  
                C[i][j] += A[i][k] * B[k][j]  
            }  
        }  
    }  
}
```

Matrix Multiplication

- What if we use the transpose to change the visit order of the matrix?

```
// ***** C = A^T x B ***** //  
void matmul_AT() {  
    memset(C, 0, sizeof(C));  
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < n; j++) {  
            AT[i][j] = A[j][i];  
        }  
    }  
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < n; j++) {  
            for (int k = 0; k < n; k++) {  
                C[i][j] += AT[k][i] * B[k][j];  
            }  
        }  
    }  
}
```

- What is the difference on hit ratio?

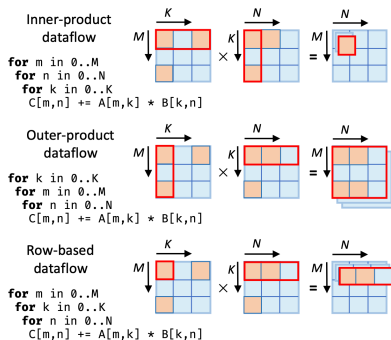
- Meanwhile, loop order (dataflow) may also affect the performance.
- Think about using additional optimization approaches to improve the hit ratio.

Matrix Multiplication

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- Think about using additional optimization approaches to improve the hit ratio.

```
void matmul_tiling() {  
    memset(C, 0, sizeof(C));  
    int iTile = 64, jTile = 64, kTile = 32;  
    for (int i_outer = 0; i_outer < n / iTile; i_outer++) {  
        // Write your code here  
    }  
}
```

Matrix Multiplication



For each element's value on output matrix C:

- Inner: read A, B for K times, write C for once
- Outer: read A, B only for once, write C for K times (by frame)
- Row-based: read A for once, read B for K times, write C for K times (by row)

THANK YOU!